MZUZU UNIVERSITY

FACULTY OF INFORMATION SCIENCE AND COMMUNICATIONS

DEPARTMENT OF INFORMATION AND COMMUNICATION TECHNOLOGY

BACHELOR OF SCIENCE IN INFORMATION AND COMMUNICATION TECHNOLOGY (BICT, BSCEICT)

SUPPLEMENTARY /DEFERRED EXAMINATION

ICT 1102 : INTRODUCTION TO PROGRAMMING

TIME ALLOWED: 3 HOURS **DATE : 14TH JUNE , 2017**

INSTRUCTIONS

i. This paper contains **TWO** pages. Please check.
ii. There are **Twelve Questions** in this paper. Please check.
iii. Answer **ALL** questions.
iv. Marks for each question are indicated.

1. Define the following terms:
   a. Compiler. **[2 marks]**
   b. Interpreter. **[2 marks]**
   c. Syntax. **[2 marks]**
   d. Reserved word. **[2 marks]**
   e. Variable. **[2 marks]**
2. Study the pseudo-code for the program below and answer the questions that follow:

```
Program start
Use variables: value OF TYPE Integer
        letter OF TYPE Character
DISPLAY "Enter Letter"
ACCEPT letter
DISPLAY "Enter a positive number"
ACCEPT value
WHILE value <> 0 DO
        PRINT letter
        value:=value – 1
ENDWHILE
DISPLAY "Thank You"
Endprogram
```

   a. Explain how the pseudo-code for the program works. **[5 marks]**
   b. What will be the output from the program if a user enters a **Q** as letter and **4** as the positive number? **[4 marks]**
3. Explain any **two** factors that can be considered when evaluating programming languages for use in software development. **[4 marks]**
4. What is the main difference between the while loop and the repeat until loop? **[4 marks]**
5. Write a pseudo code for a program that calculates and print the sum of numbers from 5 to 25. **[4 marks]**
6. Consider the following pseudo-code program segment that allows entry of a number in the range of 0 to 100.

```
Program start
Use variables: number OF TYPE Integer
REPEAT
        DISPLAY "Enter a number between 0 and 100"
        ACCEPT number
UNTIL number < 0 or number > 100
end program
```

   a. Draw a *flowchart* diagram to show this algorithm. **[3 marks]**
   b. Rewrite the pseudo-code for the program segment above using the *WHILE..DO* loop. **[4 marks]**

7. Write a pseudo-code for a program to accept a positive integer value from a user, and compute the following sequence: If the number is even, halve it; if it's odd, multiply by 3 and add 1. Repeat this process until the value is 1, printing out each value. **[6 marks]**

Typical output might be:

Initial value is 5
Next value is 16
Next value is 8
Next value is 4
Next value is 2
Final value is 1

8. Write a pseudo code for a program that calculates and print the arithmetic mean of odd numbers in the range of 0 to 21. **[6 marks]**
9. Write a pseudo code for a program that will print a series of numbers from 10 to 3 in reverse order. Use a While Loop. **[5 marks]**
10. Write pseudo code for a program that allows the entry of two decimal numbers. The program should then calculate division operation on the two numbers. Since dividing by zero is an illegal operation, the program should deny such and allow a user to enter the numbers again with a non-zero divisor. The operations should be repeated until a user enters an**E** to quit program execution. **[7 marks]**
11. Write a pseudo code for program that will calculate and print the cube of a number entered by a user until the user enters a number which is less than zero or more than 100. **[6 marks]**
12. Write a pseudo code for aprogram to determine whether the character entered at the keyboard is a *vowel*. **[5 marks]**

END OF QUESTION PAPER